

INPUT

- I1. Taglish disaster tweet corpus, four typhoon events
- I2. Four multilingual Transformer encoders
- I3. MTO configurations with STL and static baselines
- I4. Local FP16 workstation testbed

PROCESS

- P1. Normalization, tokenization, BIO alignment, LOEO splits
- P2. Multi-task fine-tuning with gradient diagnostics
- P3. Bootstrap intervals and Friedman-Nemenyi tests
- P4. FastAPI embedding with gazetteer resolution

OUTPUT

- O1. Hard-sharing multi-task model, $O(1)$ inference
- O2. Benchmark tables across 12 paired runs
- O3. Bootstrap and non-parametric results
- O4. Prototype disaster triage application